That is, the MLE equals the parameter value for which the probability of the observed data is maximised.

The probability of the data for given parameter value $p_\varphi(\boldsymbol{x})$ is also known as the likelihood function, hence the name of the MLE. When the measurement outcomes are independent of each other, the probabiities multiply $p_\varphi(\boldsymbol{x}) = p_\varphi(x_1)\cdots p_\varphi(x_N)$. Taking the logarithm then, we get a sum $\log p_\varphi(\boldsymbol{x}) = \sum_k \log p_\varphi(x_k)$. Since the logarithm is monotonically increasing, maximising $p_\varphi(\boldsymbol{x})$ is equivalent to maximising $\log p_\varphi(\boldsymbol{x})$. It is often convenient to work with $\log p_\varphi(\boldsymbol{x})$, known as the log-likelihood.

Example 4.2: Maximum-likelihood estimator under Gaussian noise

Consider N independent repetitions of a measurement subject to Gaussian noise, as in Example 4.1. What is the MLE for this measurement?

The probability for observing a given sequence of outcomes $\boldsymbol{x} = (x_1, \ldots, x_N)$ is

$$p_\varphi(\boldsymbol{x}) = p_\varphi(x_1)\cdots p_\varphi(x_N) \tag{4.14}$$

$$= (2\pi\sigma^2)^{-\frac{N}{2}} \exp\left(-\sum_{k=1}^{N} \frac{(x_k-\varphi)^2}{2\sigma^2}\right). \tag{4.15}$$

Maximising this expression with respect to φ is equivalent to minimising the sum of squares in the exponent. Hence

$$\tilde{\varphi}_{MLE}(\boldsymbol{x}) = \operatorname*{argmin}_{\varphi} \sum_{k=1}^{N} (x_k-\varphi)^2. \tag{4.16}$$

This minimum is obtained when φ equals the mean of the x_k. Thus, in this case, the MLE is just the average

$$\tilde{\varphi}_{MLE}(\boldsymbol{x}) = \frac{x_1+\cdots+x_N}{N}. \tag{4.17}$$

The expected value is $\langle\tilde{\varphi}_{MLE}(X)\rangle = \frac{\varphi+\cdots+\varphi}{N} = \varphi$. Hence, the MLE for this measurement is unbiased.

The MLE is not generally unbiased for finite samples. However, it is asymptotically unbiased for large N, up to terms og order $\frac{1}{\sqrt{N}}$, and it achieves the Cramér-Rao bound asymptotically. For particular measurements, the MLE may be unbiased for any N, as in Example 4.2.

4.1.3 The Cramér-Rao bound

Unbiased estimators

The Cramér-Rao bound[2] states that, for any unbiased estimator $\tilde{\varphi}$ corresponding to a measurement X the variance is bounded from below as

$$\langle\Delta^2\tilde{\varphi}\rangle \geq \frac{1}{\mathcal{F}_\varphi(X)}. \tag{4.18}$$

[2]Named for Harald Cramér (1893-1985) and Calyampudi Radhakrishna Rao (1920-2023).

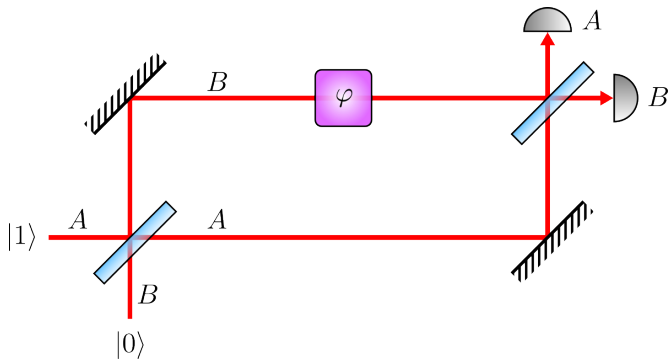
Note that, since the estimator is unbiased, the variance also equals the mean-squared error $\mathrm{MSE}(\tilde{\varphi}) = \Delta^2\tilde{\varphi}$. The Fisher information thus sets a lower bound on the best possible precision we can hope to obtain from a given measurement. An estimator which achives equality in the Cramér-Rao bound is said to be *efficient*.

Because the Fisher information is additive for independent measurements (see Prob. 4.2), if a parameter is estimated by repeating the same experiment N times, with each repetition corresponding to X, then

$$\langle\Delta^2\tilde{\varphi}\rangle \geq \frac{1}{N\mathcal{F}_\varphi(X)}. \tag{4.19}$$

The variance of the best estimate thus improves as $\frac{1}{N}$.

Example 4.3: Phase estimation with single photons



Consider the task of estimating the phase difference between two optical paths. This can be done using a single photon in a setup as shown (known as a Mach-Zender interferometer). The photon is first split on a balanced beam splitter into a superposition of two paths. One path (B) undergoes a phase shift φ while the other (A) does not. The paths are the recombined on a second beam splitter and the outputs are measured with single-photon detectors. What precision can be achieved for estimating φ with this strategy?

Assuming no loss, the state transforms as

$$|1,0\rangle_{AB} \underset{\text{first BS}}{\longrightarrow} \frac{1}{\sqrt{2}}\left(|1,0\rangle_{AB} + |0,1\rangle_{AB}\right) \tag{4.20}$$

$$\underset{\text{phase shift}}{\longrightarrow} \frac{1}{\sqrt{2}}\left(|1,0\rangle_{AB} + e^{i\varphi}|0,1\rangle_{AB}\right) \tag{4.21}$$

$$\underset{\text{second BS}}{\longrightarrow} \frac{1+e^{i\varphi}}{2}|1,0\rangle_{AB} + \frac{1-e^{i\varphi}}{2}|0,1\rangle_{AB} \tag{4.22}$$

$$= e^{i\frac{\varphi}{2}}\left[\cos\left(\frac{\varphi}{2}\right)|1,0\rangle_{AB} + i\sin\left(\frac{\varphi}{2}\right)|0,1\rangle_{AB}\right]. \tag{4.23}$$

Assuming perfect detectors, there are two possible outcomes: a single click at A or B (denoted 0 and 1, respectively) with probabilities

$$p_\varphi(0) = \cos^2(\frac{\varphi}{2}) \quad \text{and} \quad p_\varphi(1) = \sin^2(\frac{\varphi}{2}). \tag{4.24}$$

From (4.1)-(4.2), the Fisher information is

$$\mathcal{F}_\varphi = \frac{[2\cos\left(\frac{\varphi}{2}\right)\sin\left(\frac{\varphi}{2}\right)\frac{1}{2}]^2}{\cos^2(\frac{\varphi}{2})} + \frac{[-2\cos\left(\frac{\varphi}{2}\right)\sin\left(\frac{\varphi}{2}\right)\frac{1}{2}]^2}{\sin^2(\frac{\varphi}{2})} = 1. \tag{4.25}$$

The Cramér-Rao bound then stipulates that for N independent repetitions of this experiment (i.e. using N single photons independently) and any unbiased estimator $\langle\Delta^2\tilde{\varphi}\rangle \geq 1/N$. This can be improved by instead using N photons to prepare multi-photon entangled states, as discussed in Sec. 4.2 and Chap. 5.

Proof for unbiased estimators

The Cramér-Rao bound follows from the Cauchy-Schwartz inequality for random variables. The Cauchy-Schwartz inequality states that for any pair of random variables A, B

$$\mathrm{Cov}(A,B) \leq \langle\Delta^2 A\rangle\langle\Delta^2 B\rangle, \tag{4.26}$$

where $\langle\Delta^2 A\rangle = \langle(A-\langle A\rangle)^2\rangle$ denotes the variance as usual (similarly for B) and $\mathrm{Cov}(A,B) = \langle(A-\langle A\rangle)(B-\langle B\rangle)\rangle$ the covariance.

Take the random variables $A = \tilde{\varphi}(X)$ and $B = \frac{\partial}{\partial\varphi}\log(p_\varphi(X))$. Then by definition $\mathrm{Var}(A) = \Delta^2\tilde{\varphi}$ and $\mathrm{Var}(B) = \mathcal{F}_\varphi(X)$. Because $\tilde{\varphi}$ is unbiased, $\langle A\rangle = \varphi$, and because the mean score is zero, $\langle B\rangle = 0$. The covariance is therefore

$$\begin{aligned}
\mathrm{Cov}(A,B) &= \langle(\tilde{\varphi}(X)-\varphi)\frac{\partial}{\partial\varphi}\log p_\varphi(X)\rangle && (4.27)\\
&= \langle\tilde{\varphi}(X)\frac{\partial}{\partial\varphi}\log p_\varphi(X)\rangle && (4.28)\\
&= \sum_x \tilde{\varphi}(x)\frac{\partial}{\partial\varphi}p_\varphi(x) && (4.29)\\
&= \frac{\partial}{\partial\varphi}\sum_x \tilde{\varphi}(x)p_\varphi(x) && (4.30)\\
&= \frac{\partial}{\partial\varphi}\varphi = 1. && (4.31)
\end{aligned}$$

The Cauchy-Schwartz inequality then implies

$$\langle\Delta^2\tilde{\varphi}\rangle\,\mathcal{F}_\varphi(X) \geq 1, \tag{4.32}$$

and the Cramér-Rao bound follows.

Biased estimators

The Cramér-Rao bound can be generalised to estimators with nonzero bias. In this case, the mean-squared error and the variance do not coincide, c.f. Eq. (4.12). It is trivial to obtain a small variance. For example, an estimator which just outputs a constant independent of φ will have zero variance but

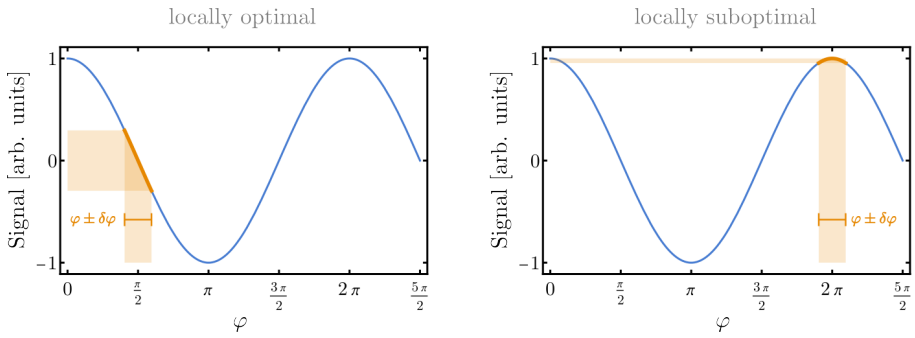


Figure 4.2: Illustration of local estimation. We consider a measurement where the outcome has a cosine dependence on the parameter φ. *Left:* If the true parameter value is close to $\frac{\pi}{2}$, where the slope of the cosine is maximal, the output will vary strongly with small changes in φ. In this case, this measurement is optimal *Right:* If the true parameter value is instead close to 2π, the signal varies little with changes in φ, and the measurement is not optimal.

is of course not a very good estimator. The mean-squared error, however, is bounded. Let $\tilde{\varphi}$ be an estimator with bias $B(\tilde{\varphi})$ and let $\dot{B}(\tilde{\varphi}) = \frac{\partial}{\partial\varphi}B(\tilde{\varphi})$ denote the derivative. Then

$$\mathrm{MSE}(\tilde{\varphi}) \geq \frac{1 + \dot{B}(\tilde{\varphi})}{\mathcal{F}(X)} + B^2(\tilde{\varphi}). \tag{4.33}$$

In the unbiased case, $B(\tilde{\varphi}) = 0$, this reduces to the standard Cramér-Rao bound.

4.1.4 Local estimation and prior knowledge

We should note that quantifying estimate precision via the Fisher information and Cramér-Rao bound means that one is taking a *local* approach to estimation. A particular estimation protocol (e.g. a particular choice of measurement) may be optimal for certain parameter ranges and not for others, as illustrated in Fig. 4.2. Furthermore, the optimal estimator, saturating (achieving equality in) the Cramér-Rao bound, may also depend on the parameter value φ. This may seem contradictory as picking the right estimator then requires that the parameter is known in advance, which makes estimating it rather pointless! What it means in practice is that the bounds derived in this manner are relevant to situations where one has good prior knowledge of the parameter value and is interested in detecting small deviations from this value. There are many appications where this is the case. Furthermore, in the regime where a lot of data is collected, one can always use a small fraction of data to constrain the parameter value and then employ an optimal local estimator on the remaining data. In the asymptotic limit of large samples, $N \to \infty$, the Cramér-Rao bound can always be saturated and in some cases it can also be attained for finite N.

In cases where the prior knowledge of the parameter is not sufficiently tight and the sample cannot be considered asymptotically large, it is better to adopt Bayesian approach to estimation. Here, the prior knowledge is encoded in a